Concern Resolution Process

QC Story

QC STORY PROJECT

PROJECT TITLE :- To eliminate damage at C- pillar area in Double cab Scorpio BIW

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio Body shop.

	Plan	Actual
Start Date	**22/07/12**	**23/07/12**
Completion Date	**31/10/12**	**20/10/12**

Current Status: Completed

Phase II – QC Story

- Team Information
- Step1: Select the Theme
- Step 2: Justify the choice
- Step 3: Understand the current situation
- Step 4: Select Targets
- Step 5: Analysis (Level 1 Analysis)
- Step 6: Implement corrective measures
- Step 7: Confirm the Effects
- Step 8: Standardize
- Step 9: Summarise & Plan future actions
- Guidelines to user

Cross Functional Team

Team Leader – Leads the team, Co-ordinates the meeting
Mentor - Facilitates the team for concern resolution, provides resource.
Mostly Mentor is PU Manager.
Members – Cross Functional Team

No	Description	Name	Designation	Dept	Phone	Email
1	**Team Leader**	Sushil Dwivedi	Assistant Manager	SBS (Tooling)	9922971514	dwivedi.sushil2@mahindra.com
2	**Mentor**	Ashutosh Hasme	Manager	Tooling	9822248250	hasme.ashutosh@mahindra.com
3	**Member 1**	Dinesh Jangam	Manager	Production	9850848982	jangam.dinesh@mahindra.com
4	**Member 2**	Nitin Hiray	Deputy Manager	MQA	9881737077	hiray.nitin@mahindra.com
5	**Member 3**	Sahdeo Masurkar	Assistant Manager	Maintnce	9766140001	masurkar.sahdeo@mahindra.com

Cross Functional Team

No	Description	Name	Designation	Dept	Phone	Email
6	**Member 4**	Nitin Pawar	Manager	Supplier	8380021964	PAWAR.NITIN@mahindra.com
7	**Member 5**	Debebrata Giri	Assistant Manager	SQA	9767744812	GIRI.DEBEBRATA@mahindra.com

1. Select the Theme

Brief description of Problem

What- To eliminate damage of C-pillar in Double cab Scorpio BIW.
Where – Major damage at side Assy.
When – During Buy-off
Who – Buy-off operator & check-man
How – 37 concerns reported in F-12 FTBO and 19 cases up to June -2012

1. Select the Theme

Brief description of Problem

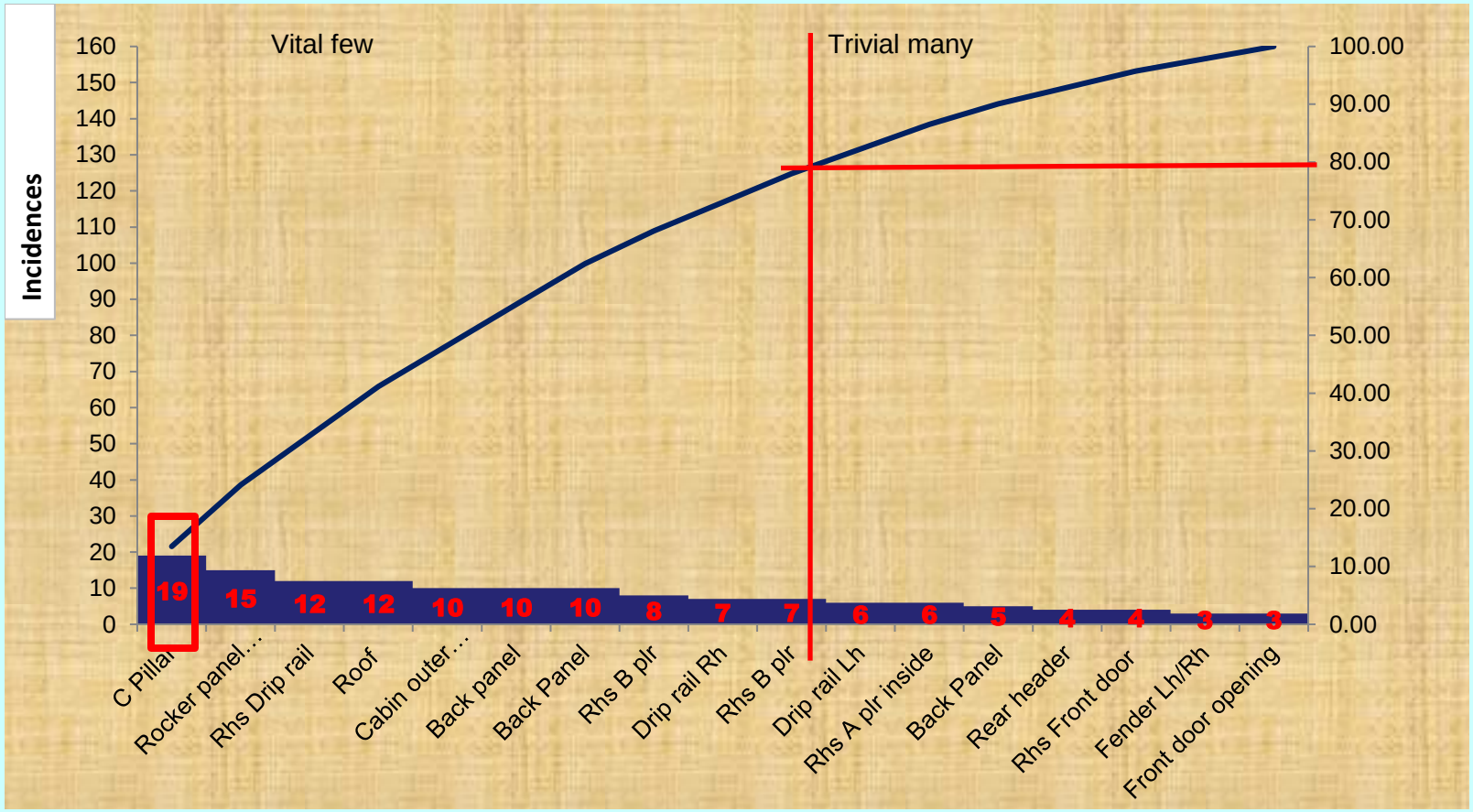
37 concerns reported in F-12 FTBO and 19 cases up to June -2012

Rework cost of C-pillar damage in F-12 FTBO is 30x672x37/60 =12432 Rs. And Rework cost of C-pillar in F-13 up to june-2012 is 30x672x19/60 = 6384 RS.

Damaged points in C-pillar during Assy

2. Justify the choice (reasons)

Pareto for defects of C- pillar damage in double cab Scorpio for month of April-June 20112 justifying the choice



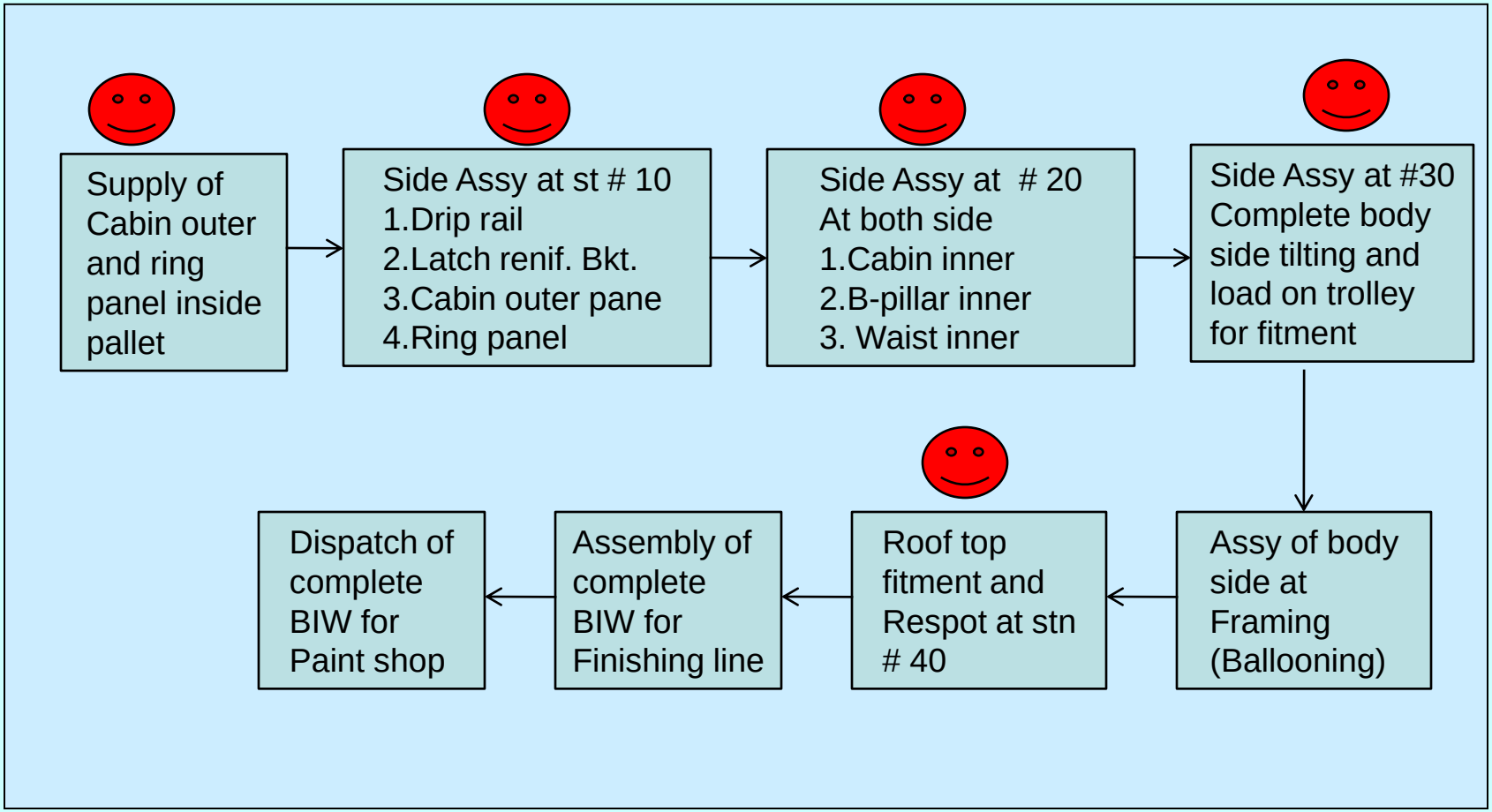
2.a Immediate Containment Action

- Initiated 100 % check at side buy off.
- Initiated 100 % Rework at body in buyoff .

C- Pillar Damage in Double cab Scorpio BIW

3. Understand the current situation

Observation of process



Defect prone area

3. Understand the current situation

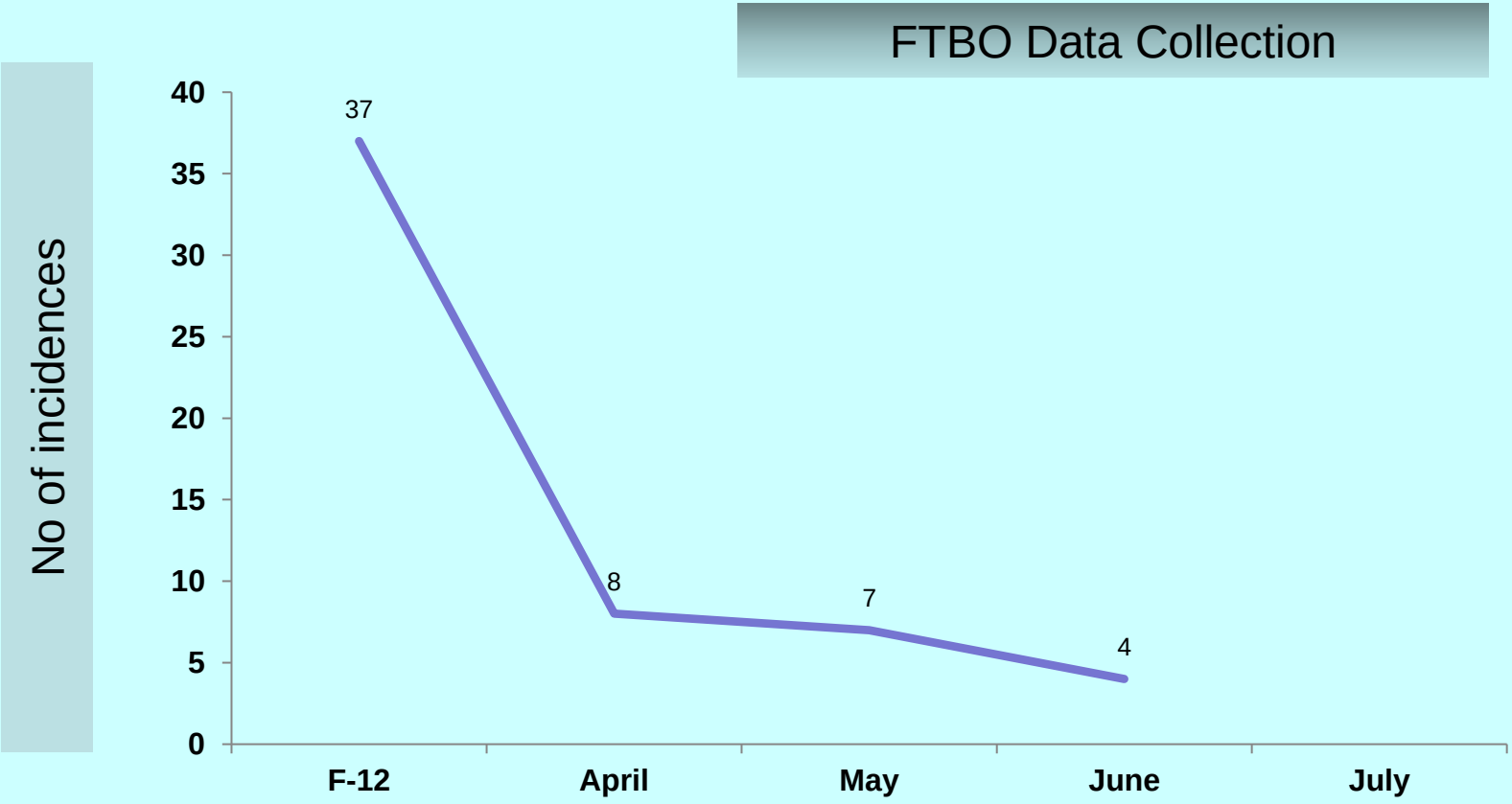
Observation of process

C-pillar damage has been observed All over the place of cabin outer and Ring panel as par mentioned area in picture.

Irrespective of shift and operator skill level the concern persists in all the shifts erratically.

3. Understand the current situation

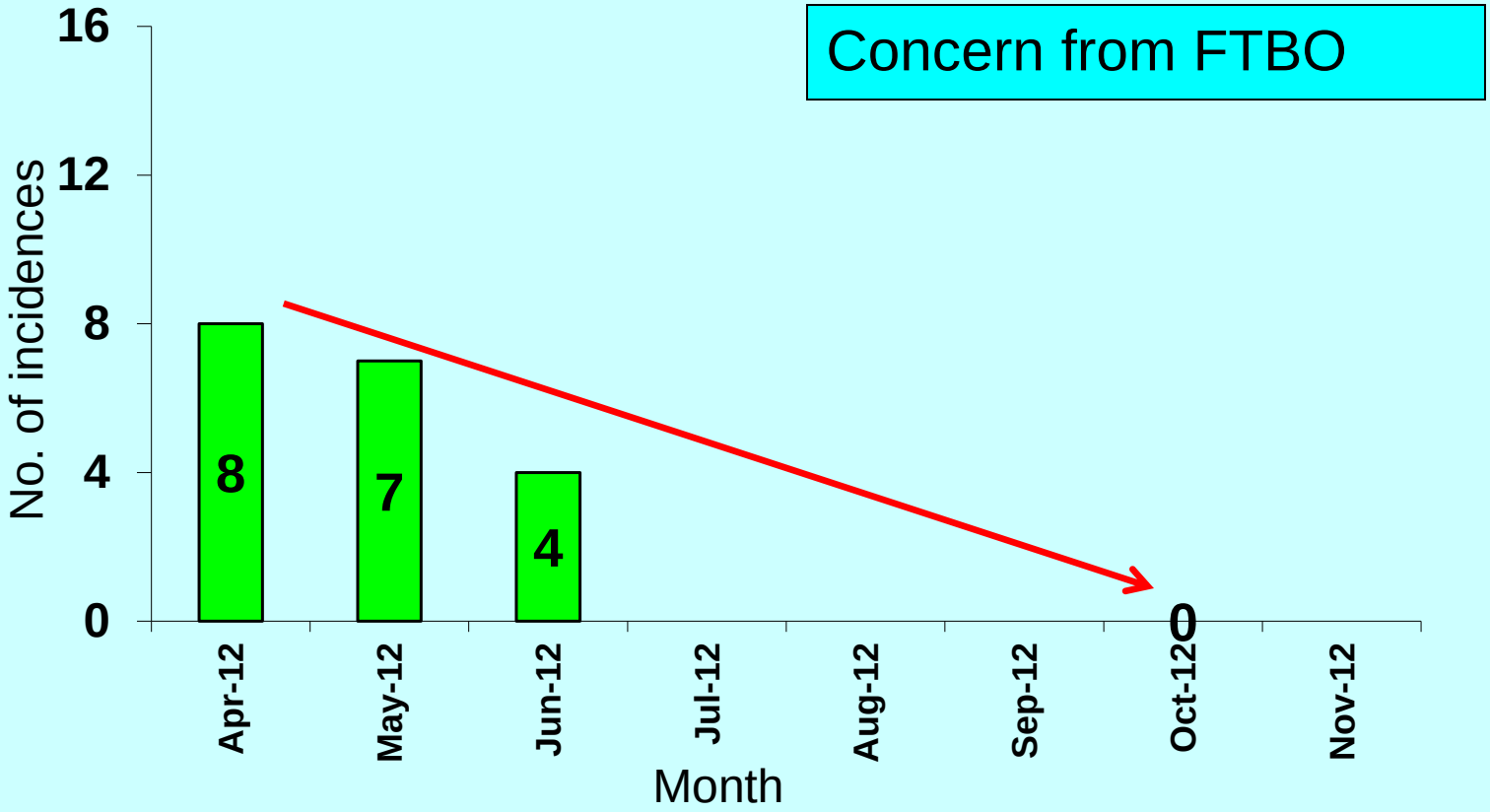
Observation of process



4. Select Target

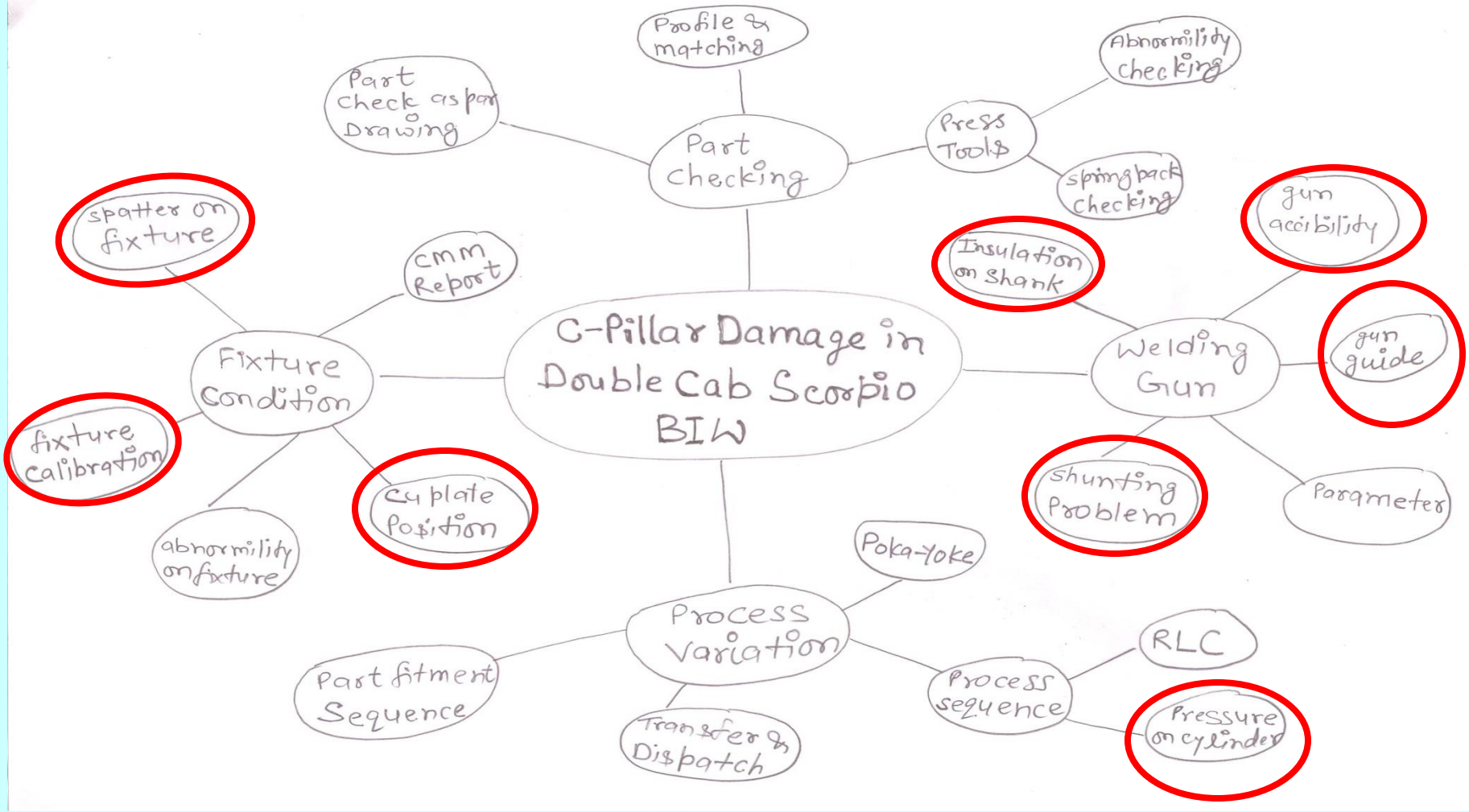
Present Level: 19 cases reported in FTBO from April to June 2012.

Target: Elimination of the concern till Oct 2012.



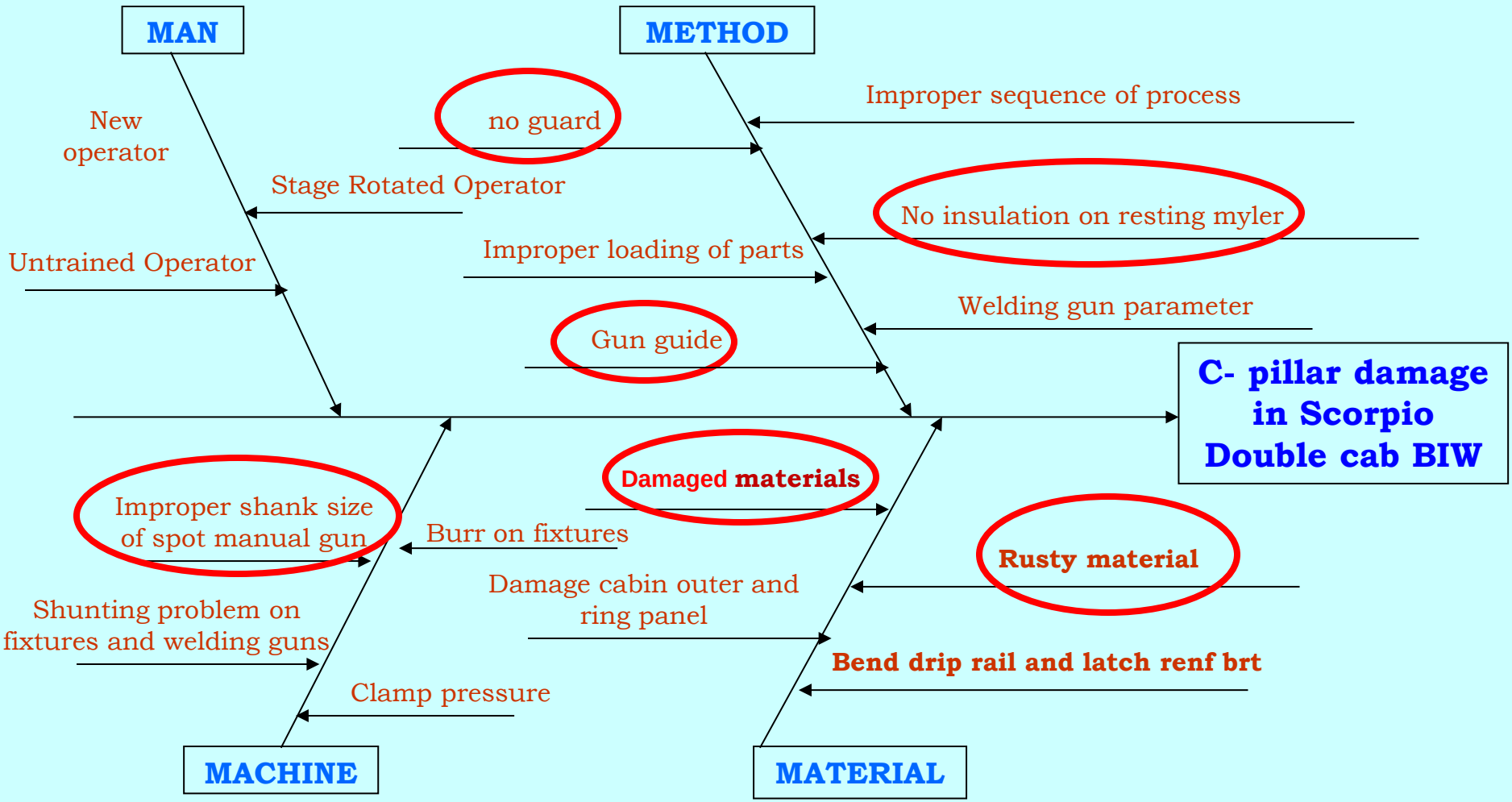
5. Analysis

5.1 Brainstorming by CFT:



5. Analysis

5.1 Identify possible causes:



TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
1	C-pillar damage at flange during Assy	Gun guard checked on line for C-pillar at side Assy #20.	Gun guard was not available at side Assy #20 near window.	valid
2	C-pillar damage at upper side	Fixture condition checked at upper side of c-pillar with panel.	Gun guard was worn out at side Assy #10.	valid
3	Process Sequence	Process sequence of manufacturing checked and ensure by operators.	We observed right process sequence at side Assy #10 and # 20.	Invalid
4	Shunting of spot gun with C- pillar	We checked the shunting insulation on welding gun and resting mylers at side Assy #10 and #20.	No permanent insulation was available in spot welding gun holder and resting mylers	valid
5	Spot taken at wrong position of c-pillar.	We checked the gun shank at side Assy #10 and #20 near C- pillar, where get damage.	Observed gun shank length was more, due to this c-pillar damaged many times.	valid
6	Parts Fitment on line.	Checked parts fitment sequence and care on line at side Assy #10 and #20 both side.	We observed on line at side Assy #10 and #20 both side LH and RH that part fitment sequence is correct as par MAPS SOS.	Invalid
7	C-pillar damage near cu plate	We checked Cu plate condition mtg style and floating at side Assy both LH and RH.	Cu plate mtg was not proper and provided only one mtg, due to this c-pillar damaged many times.	valid

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
1	C-pillar damage at flange area during Assy	Gun guard checked on line for C- pillar at side Assy #20.	Gun guard was not available at side Assy #20 near window.	valid

No gun guard is available near window area at c-pillar

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
2	C-pillar damage at upper side	Fixture condition checked at upper side of c-pillar with panel.	Gun guard was worn out at side Assy #10.	valid

Gun guide worn out at side Assy #10

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
3	Process Sequence	Process sequence of manufacturing checked and ensure by operators.	We observed right process sequence at side Assy #10 and # 20.	Invalid

Process sequence is correct as par SOS as shown in picture

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
4	Shunting of spot gun with C- pillar	We checked the shunting insulation on welding gun and resting mylers at side Assy #10 and #20.	No permanent insulation was available in spot welding gun holder and resting mylers	valid

Resting myler and welding gun has no insulation

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
5	Spot taken at wrong position of c-pillar.	We checked the gun shank at side Assy #10 and #20 near C- pillar, where get damage.	Observed gun shank length was more, due to this c-pillar damaged many times.	valid

Shank length is more as shown in picture

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
6	Parts Fitment on line.	Checked parts fitment sequence and care on line at side Assy #10 and #20 both side.	We observed on line at side Assy #10 and #20 both side LH and RH that part fitment sequence is correct as par MAPS SOS.	Invalid

Part fitment is correct as par SOS

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
7	C-pillar damage near cu plate	We checked Cu plate condition mtg style and floating at side Assy both LH and RH.	Cu plate mtg was not proper and provided only one mtg, due to this c-pillar damaged many times.	valid

Cu plate mtg is single, due to this movement of Cu plate may be possible in any direction

WHY - WHY Analysis

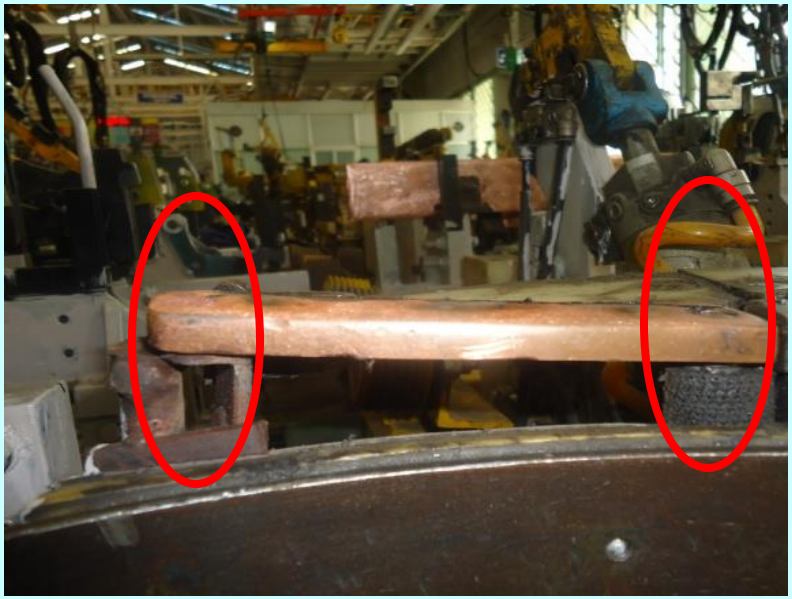
5.3 Identify the root causes: (Use Format MQS/MAPS/F05)

Valid Probable Cause	WHY	WHY	WHY	WHY	ACTION
C-pillar damage at flange area during Assy	**Spot taken at upper side of c-pillar**	**Guard availability**			**Guard Made at c-pillar upper area with support to panel.**
C-pillar damage at upper side	**Concern created i.e. C- pillar damage**	**Gun guide availability**	**Accessibility problem**		**Accessibility created and gun guide provided**
Shunting of spot gun with C- pillar	**Spot taken with touching of resting mylers**	**No insulation**			**Insulation provided on welding gun shank and resting mylers.**
Spot taken at wrong position of c-pillar.	**Shank length is more**	**Wrong design**			**Shank design changed and length increased.**
C-pillar damage near cu plate	**Movement of Cu plate**	**Single mtg of Cu plate**			**Double mtg provided on Cu plate at body side Assy #10.**

6. Implement the corrective measures

Root Cause	Action Plan	Target Dt	Resp.	Status
Cu plate mtg	Cu plate fixed with both end and float provided also(LH/RH).	17/08/12	Sushil	Closed

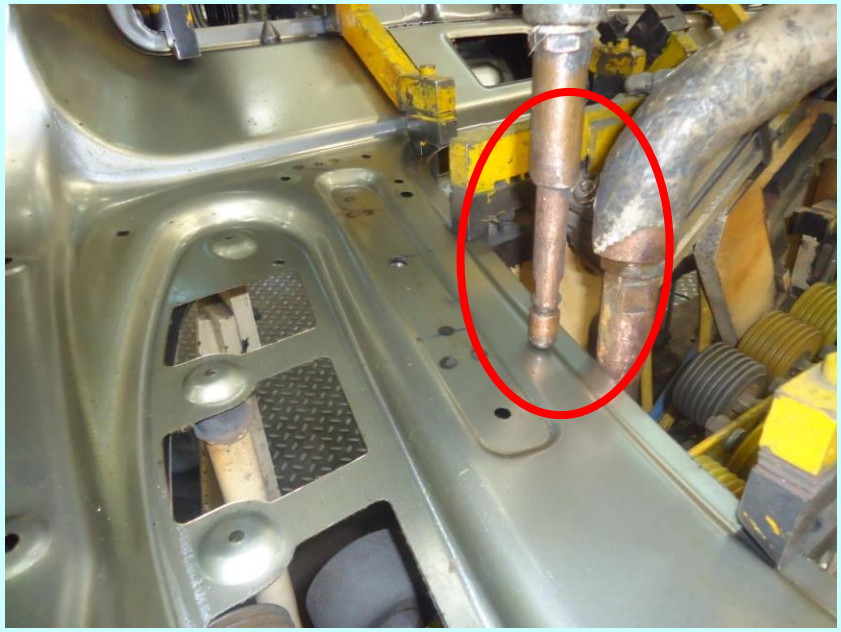
Single Cu plate mounting at W/H area on side Assy #10 both side



Double mounting of Cu plate at W/H on side Assy #10 both side

6. Implement the corrective measures

Root Cause	Action Plan	Target Dt	Resp.	Status
Guard at side Assy	Guard Provided at upper side of c-pillar both side(LH/RH).	24/08/12	Sushil	Closed



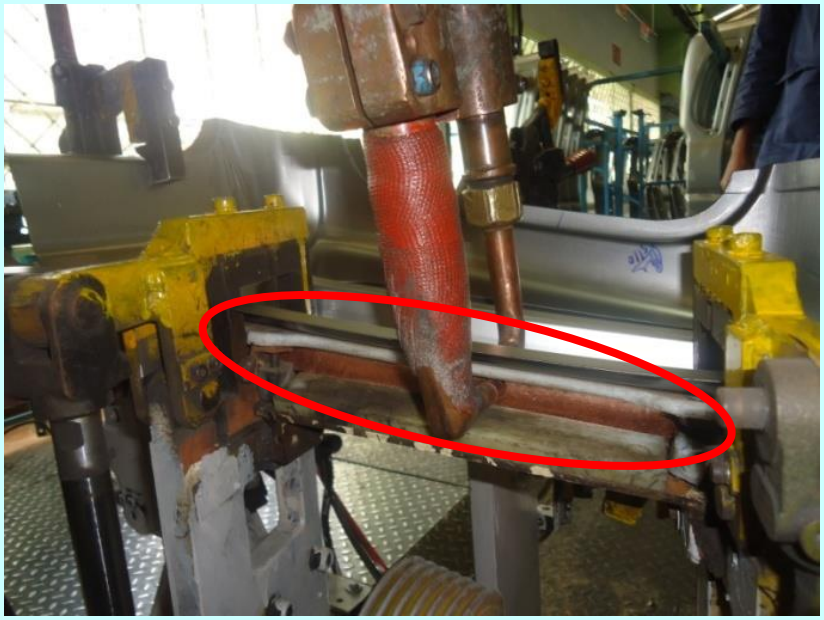
No guard at C-pillar near window area at side Assy # 20

New nylon guard provided at side Assy #20 both side near window area

6. Implement the corrective measures

Root Cause	Action Plan	Target Dt	Resp.	Status
Gun guide for C-pillar	Gun guide provided at C-pillar for welding of cabin inner and cabin outer.	21/09/12	Sushil	Closed

Nylon Gun guard worn out at side Assy #10 both side LH/RH

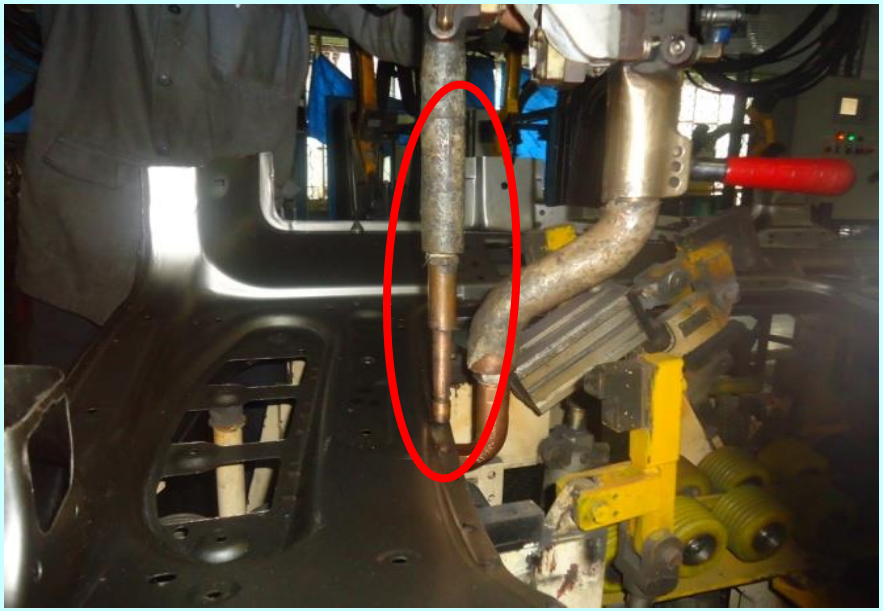


New nylon guard made at side Assy #10 both LH/RH

6. Implement the corrective measures

Root Cause	Action Plan	Target Dt	Resp.	Status
Welding gun shank design	Welding gun shank length increased.	23/09/12	Sahdeo	Closed

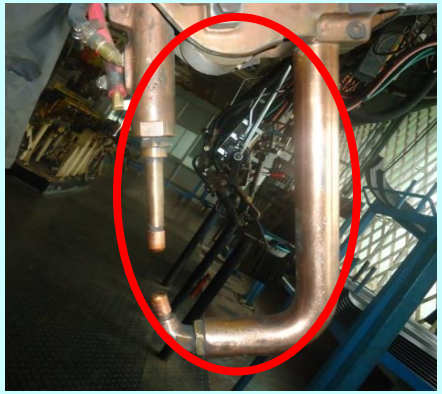
Shank length is less as par requirement at side Assy #20



Shank length increased as par our requirement at side Assy #20

6. Implement the corrective measures

Root Cause	Action Plan	Target Dt	Resp.	Status
Shunting of spot gun with C- pillar	Insulation provided at welding gun and resting mylers.	21/09/12	Sushil	Closed



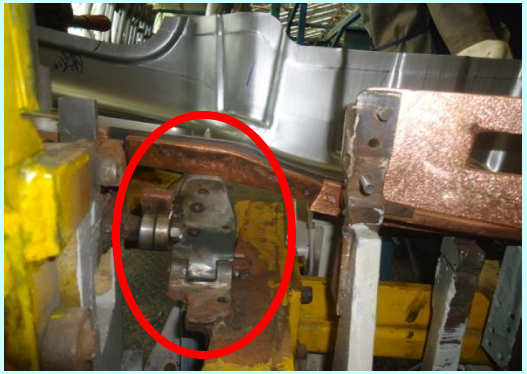
Shunting problem in welding guns and resting mylers at side Assy #10 & #20

Insulation provided in welding gun shank and resting mylers at side Assy #10& #20

Some other Pro-active actions

C-pillar may damage due to gun inserting more between mylers at #20, so nylon guard provided.

C-pillar may damage due to gun inserting more between pillars near window area at #10, so MS guard provided.

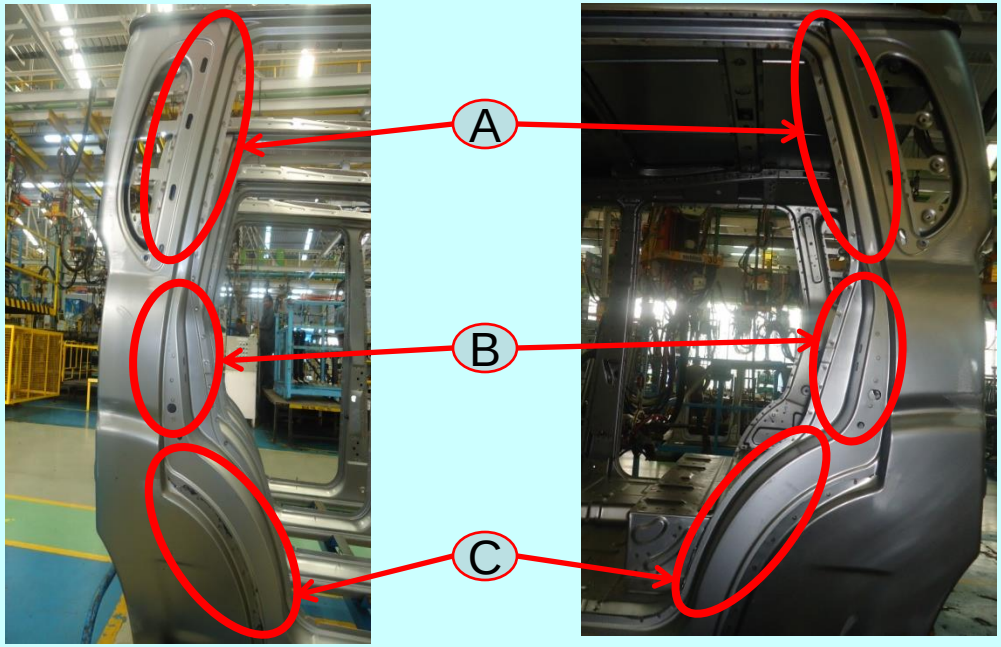


C-pillar may damage due to taken spot between two Cu plate edges, so single cu plate made at side Assy #10.

7. Check & confirm the results & effects

7.1 Verify Results:

After implementation of corrective measures at body side Assy station and framing line in body shop, it was observed Damages are eliminated. Following were the readings of damages area wise :-



Body side Assy RH

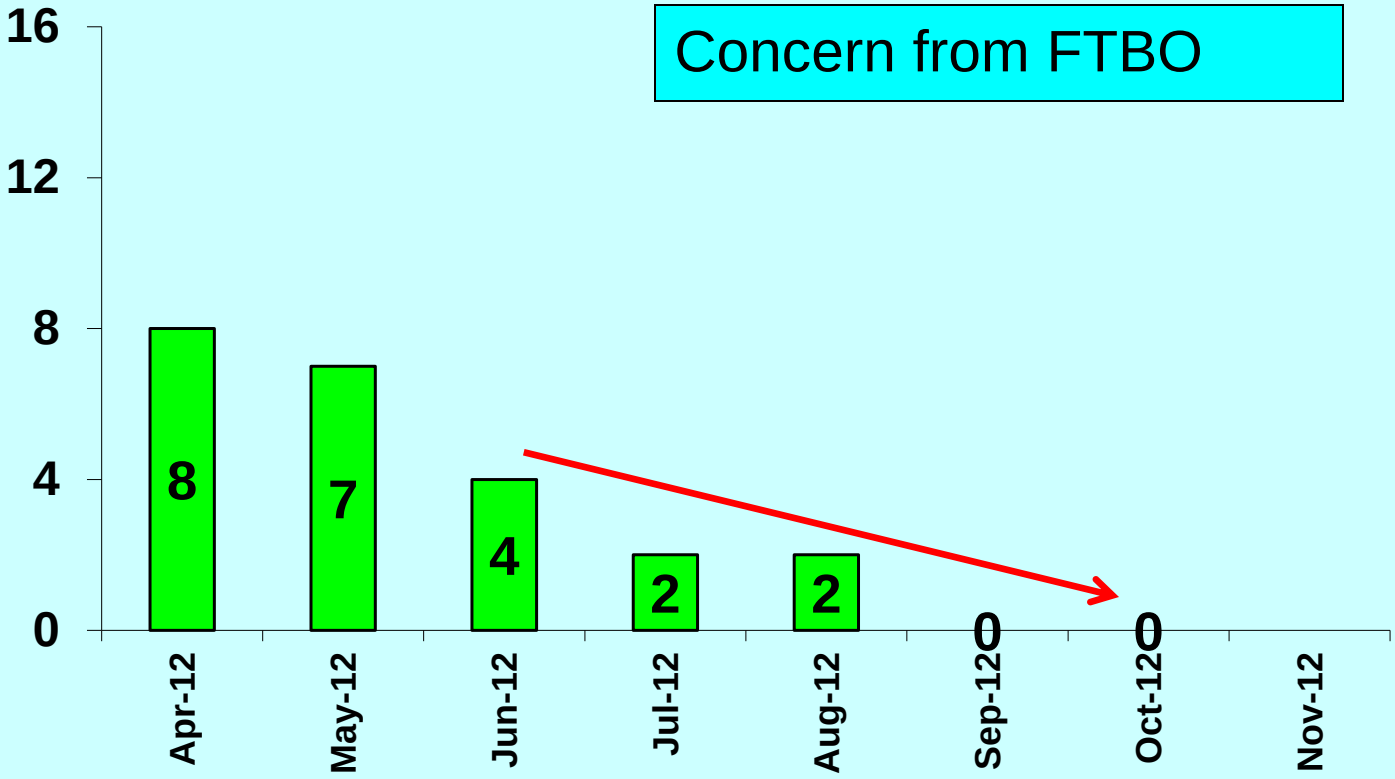
Body side Assy LH

Location.	May 12	Jun 12	Jul 12	Aug 12	Sep 12	Oct 12 Till
A	4	2	1	1	0	0
B	1	1	1	0	0	0
C	2	1	0	1	0	0

7. Check & confirm the results & effects

7.1 Verify Results:

After implementation of corrective measures at body side Assy in Scorpio body, the concern of C- pillar damage reduced in the month of Aug- Sept 2012 so far as reflected in trend shown below,



7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

Change implemented 100% from 01- Oct 2012.

Root Cause	Implementation Plan	Target Dt	Resp.	Status
Cu plate mtg	Cu plate fixed with both end and float provided also(LH/RH).	17/08/12	Sushil	Closed
Guard at side Assy	Guard Provided at upper side of c-pillar both side(LH/RH).	24/08/12	Sushil	Closed
Gun guide for C-pillar	Gun guide provided at C-pillar for welding of cabin inner and cabin outer.	21/09/12	Sushil	Closed
Shunting of spot gun with C- pillar	Insulation provided at welding gun and resting mylers.	21/09/12	Sushil	Closed
Welding gun shank design	Welding gun shank length increased.	23/09/12	Sahdeo	Closed

Check & confirm the results & effects (Cont……)

7.3 Tangible & Intangible benefits:

1. **FTBO improved 3% at Buy- off and TCF ID improved 9 unit also.**
2. **Reduction in rework cost. (Appx. Rs 12432 /- per anum)**
3. **Operator moral improved due to elimination of C-pillar irritating rework at Buy-off.**

8. Standardisation

8.1 Updation of documents:

SOS updated,

Fixture history card updated, which is attached.

Guard checklist updated, which is attached.

PFMEA updated, which is linked with Slides

Control plan updated, which is linked with slides.

8.2 Training:

Operators trained at side Assy for detection of C- pillar damage, Spot welding Gun, final damage at the final stage.

8.3 Additional Audit / Check

1. Check point added in officer checklist.
2. Daily process audit for chronic concern started.

9. Summarise & Plan future actions

9.1 Conclusion:

Failure mode to be analyze prior to gun shank design as well as design of fixture for shunting problem.

9.2 Horizontal Deployment:

Shunting insulation implemented on other area as well as nylon guard made for gun guide in framing, front floor Assy and under body area also.

9.3 Lessons Learnt:

Improper fixture and welding gun as well as gun guide leads to create damages on body Side Assy.

Thank You

Fixture history card

FIXTURE HISTORY CARD				
ZONE:			DEPT : MFG ENGG.-SCORPIO SC-DC BODY	
FIXTURE NO.		ASSY NAME:RH SIDE STN20	FREQUENCY : 6 Months and/or whenever changes done	
DATE	UNIT NO	CONCERNS	CHANGES DONE/ ACTION TAKEN	REMARKS
10/4/2007		Delta spot missing at B pillar due to mettalic block sparking	For locating and clamping operations one combined pneumatic cylinder provided	
22/11/07		Trouble to operator at the time of locating pin	Hand grip provided	
24/11/07		Spot damage - B plr	Copper plate width increased by 10 mm.	
27/12/07		Spot damage - A plr inner	Copper plate width increased by 5 mm.	
8/6/2008		Spot damage - B plr	MS Guard provided	
14/07/08		Nut off B pillar seat belt	Two guide bar provided opposite to each other to arrest DOF of assly	
25/7/2008		B Pillar seat belt nut off.	B' pillar inner shifted in +1mm (Z Direction)	
30/9/2010		Spot Damage - LH B pillar	Nylon Guard provided.	QCRT NCA3P69/10
15.10.10		Gun shunting while takeing A plr hinge spot	Nylon Guard provided.	Tear down
22/10/10		Flange Mismatch - Rhs Rear header to cabin outer DC	Cabin inner position corrected by 1.5mm in +Z Direction	QCRT NCA3P75/10
20/11/10		Flange Mismatch - Rhs Rear header to cabin outer SC	Cabin inner position corrected by 1.0mm in +Z Direction	Dock Audit
15/11/10		Fouling between Rhs Roof bow mrt.bkt to roof	Cantrail position corrected 1 mm in Z-direction at RH side assy to avoid fouling	Nova-B concern
15/12/10		Flange mismatch on Rhs Front door opening at A- plr	Clamp corrected at RH side assy.	Nova-B concern
30/12/10		Spot damage B Pillar	Stopper to be provided	FTBO Dec-10
15/1/11		Cabin outer got damage many times	Driprail clamps modifed from manual clamps to pneumatic clamping	i4-287493
22/1/11		Spot shifting from #10 to #20	A plr cu plate provided 8 mm	
6/4/2011		Damage RHS A piller inside	Guard reinf provided	FTBO
23/6/11		Flange Open (Spot damage) at A plr Drip rail area	MS guard provided on Cu plate mtg	QCRT-NCA3P0094/11
16/12/11		Damage B plr RH near latch area	Cu plate mtg modified with screw & additional washer	NCA 3P 119/11
29/05/12		B- piller RH damage	Lifter with both side rollers installed	NCA 3 P 0133
23/08/12		C- pillar damage near window area	Guard provided at C- pillar near window	FRC
SPU/ME/F/02 , Rev. 1.00,				

Fixture history card

FIXTURE HISTORY CARD				
ZONE:			DEPT : MFG ENGG.-SCORPIO SC-DC BODY	
FIXTURE NO.		ASSY NAME : LH SIDE STN20	FREQUENCY : 6 Months and/or whenever changes done	
DATE	UNIT NO	CONCERNS	CHANGES DONE/ ACTION TAKEN	REMARKS
10/4/2007		Delta spot missing at B pillar due to mettalic block sparking	For locating and clamping operations one combined pneumatic cylinder provided	
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14/07/08		Nut off B pillar seat belt	Two guide bar provided opposite to each other to arrest DOF of assly	
25/7/2008		B Pillar seat belt nut off.	B' pillar inner shifted in +1mm (Z Direction)	
30/9/2010		Spot Damage - LH B pillar	Nylon Guard provided.	QCRT NCA3P69/10
15.10.10		Gun shunting while takeing A plr hinge spot	Nylon Guard provided.	Tear down
22/10/10		Flange Mismatch - Rhs Rear header to cabin outer DC	Cabin inner position corrected by 1.5mm in +Z Direction	QCRT NCA3P75/10
20/11/10		Flange Mismatch - Rhs Rear header to cabin outer SC	Cabin inner position corrected by 1.0mm in +Z Direction	Dock Audit
15/11/10		Fouling between Rhs Roof bow mrt.bkt to roof	Cantrail position corrected 1 mm in Z-direction at RH side assy to avoid fouling	Nova-B concern
15/12/10		Flange mismatch on Rhs Front door opening at A- plr	Clamp corrected at RH side assy.	Nova-B concern
30/12/10		Spot damage B Pillar	Stopper to be provided	FTBO Dec-10
15/1/11		Cabin outer got damage many times	Driprail clamps modifed from manual clamps to pneumatic clamping	i4-287493
22/1/11		Spot shifting from #10 to #20	A plr cu plate provided 8 mm	
6/4/2011		Damage RHS A piller inside	Guard reinf provided	FTBO
23/6/11		Flange Open (Spot damage) at A plr Drip rail area	MS guard provided on Cu plate mtg	QCRT-NCA3P0094/11
16/12/11		Damage B plr RH near latch area	Cu plate mtg modified with screw & additional washer	NCA 3P 119/11
29/05/12		B- piller RH damage	Lifter with both side rollers installed	NCA 3 P 0133
25/08/12		C- pillar damage near window area	Guard provided at C- pillar near window	FRC
SPU/ME/F/02 , Rev. 1.00,				

Guard Check List

Side Assembly Guard Checking Plan							
Sr No	**Assy**	**Location (RH SIDE)**	**Week**			**Month -**	
			Week 1	**Week 2**	**Week 3**	**Week 4**	**Week 5**
1	**Side Assy #10**	A Pillar Top (Drip Rail Guard)					
		Drip Rail Guard					
		Drip Rail Guard (Above B Pillar)					
		Drip Rail Guard					
		Drip Rail Guard (Above C Pillar)					
		Drip Rail Guard					
		Drip Rail Guard(Above Cabin Outer)					
		Drip Rail Guard					
		Wheel Arch Guard (Top)					
		Wheel Arch Guard (Bottom)					
		C Pillar Guard					
		C Pillar Guard					
		C Pillar Guard					
		C Pillar Guard					
		C Pillar Guard					
		C Pillar Guard					
		C Pillar Guard					
2	**Side Assy #20**	B Pillar					
		C- pillar guard 1.10					

(SPU/ME/F/12) Rev-1.10 Effective Date -23/08/12

Guard Check List

Side Assembly Guard Checking Plan							
Sr No	**Assy**	**Location (LH SIDE)**	**Week**			**Month -**	
			Week 1	**Week 2**	**Week 3**	**Week 4**	**Week 5**
1	**Side Assy #10**	A Pillar Top (Drip Rail Guard)					
		Drip Rail Guard					
		Drip Rail Guard (Above B Pillar)					
		Drip Rail Guard					
		Drip Rail Guard (Above C Pillar)					
		Drip Rail Guard					
		Drip Rail Guard(Above Cabin Outer)					
		Drip Rail Guard					
		Wheel Arch Guard (Top)					
		Wheel Arch Guard (Bottom)					
		C Pillar Guard					
		C Pillar Guard					
		C Pillar Guard					
		C Pillar Guard					
		C Pillar Guard					
		C Pillar Guard					
		C Pillar Guard					
2	**Side Assy #20**	B Pillar					
		C- pillar guard 1.10					

(SPU/ME/F/12) Rev-1.10 Effective Date -23/08/12

PFMEA

Plant : Nasik Plant I			Model :Scorpio SC				Mahindra & Mahindra Ltd.								Cross Functional Team						
Part Nub.: 0101EL0010H			Part Name/ Process: side assy												Vivek Dhal - TOOLING		Nilin Hiraj - QA				
Latest Change Letter:			Opts No.: 10 & 20				Process Potential Failure Mode & Effect A								S Varade , CME		Jangam Dinesh - PROD				
PW Shop :Scorpio SC/DC body shop			Opts Descpts: Spot welding				Reviewed by :Lakha Pravin			Rev. No : 1.37				Doc No.: SRSD/QA/R/01	Manoharkar Sakshi - MAINT						
Modular Cell :Side assy			M/C Name & No.:Assy Fix Side				Approved by : Arbutnak Harma			Rev. Date :30.08.2012				FMEA Date (Orig) :15/10/2005							
Process Step / Function	Requirements	Potential Failure Mode	Potential Effect(s) of Failures	Sev	Class	Potential Cause(s)/Mechanism (s) of Failures	Current Process Controls Prevention	Occur	Current Process Controls Detection	Detect	RPN	Recommended Action(s)	Responsibility & Target Completion Date	Action Result - Action Taken	S	O	D	RPN			
Z 02 stn # 10 Spot weld the following components Assy plr A upper nut Assy drip rail plr A Assy chal drip rd Assy pal cabin nut Assy striker Rlf B,C Assy rlf plr A Assy Rlf plr B Fender mtg brkt Ring Panel	No spot dent/Damage C Plr curvature area (HCA3P 101/11)	Spot dent,Damage at C Pillar curvature area	Mfg - Rework at buy off (3) Assy. TCF:- Poor appearance (4)	4		Cu Plate not matching with Panel profile , Cu plate mtg to be modified.	Fixture PM	7	Visual inspection	8	224	NONE									
	Free from Spot weak - A plr drip rail (HCA3P 102/11)	Spot weak - A plr drip rail	Mfg - Rework at buy off (3) Assy. TCF:- Poor appearance (4)	4		1) Cu Plate not matching with Panel profile 2) Gun Tip alignment not ok.	1) Fixture PM 2) Operator awareness	5	Visual Inspection	1	160	NONE									
	No Damage Striker plate	Damage striker plate	Assy TCF - Striker mounting Difficult (5)	5		Gun touches to the part	operator awareness	3	Visual Inspection	1	120	NONE									
	NVH stiffener present	NVH stiffener missing	End user - Noise during vehicle running	5		Operator error	operator awareness	3	Visual Inspection	8	120	NONE									
	No spot dent /damage at W/Arch area (DC) (HCA3P 111/11)	Spot damage at W/Arch area	Mfg - Rework at buy off (3) Assy. TCF:- Poor appearance (4)	4		Cu Plate not matching with Panel profile	Fixture PM	3	Visual inspection	8	96	NONE									
Z 02 stn # 30 Collect the assy.	No damage/bend drip rail at lower end	Damage/Bend Drip rail	Mfg - Rework at buy off (3) Assy. TCF:- Fouling door to Drip rail (4)	4		1) Roller worn out 2) Fixture sharp edges	Fixture PM	3	Visual inspection	8	96	NONE									

Rev No.:	Rev Date	Revision Summary	Reason for Revision
1.01	15.04.2006	Overall DCP review	Scheduled 6 monthly review
1.31	25.06.2011	L3 QCRT concern no.91&94 added	QCRT Concern
1.35	18.10.2011	Overall DCP review	Scheduled 6 monthly review
1.36	17.4.2012	Overall DCP review	Scheduled 6 monthly review
1.37	28.09.2012	Overall DCP review	Scheduled 6 monthly review

Control plan

Plant : Nasik		Model :SCORPIO DC		Mahindra & Mahindra Ltd. Control Plan								Cross Functional Team	
Part No/s.: 0101AL0100N		Part Name/ Process:SIDE ASSY STN # 10/20										Vivek Bhat - TOOLING	Nitin Hiray - QA
Latest Change Letter:		Doc No.: SRSD/QA/R/05		Prototype : ☐		Prelaunch: ☐		Production: ■		Critical Char: <CC>		S Varade, CME	Jangam Dinesh
Aggregate No.: 0101AL0100N		Aggregate Part Descrptn: Body assy less door		Module Name : SIDE ASSY 10		Reviewed by : Lakke Pravin		Rev. No : 1.37		Significant Char : <SC>		Masurkar Sahdev- MAINT	
PU/ Shop : Scorpio SC/DC Body Shop		Next Aggregate No.: 0101AL0030N		Cell Name :SIDE ASSY		Approved by : Mr. Sandeep Deshpande		Rev. Date : 30.08.2012					CP Date(Orig) :25/10/0:
Opn No.	Operation Description	Machine, Device, Jig, Tools Description	Characteristics No.	Characteristics Product	Characteristics Process	Sp. Char a. Clas	Method Product/ Process Specification.	Evaluation Measurement Tech.	Sample Size	Sample Freq.	Control Method Record	Control Method Resp.	Reactio
10	Z 02 stn # 10 Spot weld the following	**Refer Generic contol plan for spot weld sealer application**											
		Refer Generic contol plan for spot welding											
20	Z 02 stn # 20 Spot weld the following components A pillar inner		Refer Generic contol plan for structural sealer application.										
			Refer Generic contol plan for spot welding (Delta Spot)										
			Refer Generic contol plan for spot welding										
30	**Z 02 stn # 30 Collect the assy**		3.02	Damage(Bend) Drip rail at lower side **(L2 QCRT Concern 304)**	--		Free from Damage/bend - A plr drip rail	Visual	1	Every assy.	NA	Operator	If not ok, inform
					Roller,Fixture sharp edges		Fixture sharp edges to be removed	Visual	1	Bi-Monthly	SPU/ME/R/14	Mfgg. Engg..	As per Plan / R

Rev No.	Rev Date	Revision Summary	Reason for Revision
1.01	15.04.2006	Overall DCP review	Scheduled 6 monthly review
1.02	15.11.2006	Updation of action taken for high RPN, overall DCP review	Action taken for high RPN, scheduled 6 monthly review
1.03	26.05.2007	Overall DCP review - CAPA actions updated	Scheduled 6 monthly review
1.23	27.08.2010	NVH siffner pad modification - No Change in CP & QCRT Concern updated	EN 10 -6689 N implementation - no change in Control Plan & QCRT concern no.63 updated
1.24	06.09.2010	Material change -Existing to GA A pillar drip rail LH + RH	EN10-05515 N implementation - no change in Control Plan
1.34	08.09.2011	Deletion of 4 nuts from wing brackets which required only for petrol models	EN 12-08607N implementation - no change in Control Plan
1.35	18.10.2011	Overall DCP review	Scheduled 6 monthly review
1.36	17.04.2012	Overall DCP review	Scheduled 6 monthly review
1.37	28.09.2012	Overall DCP review	Scheduled 6 monthly review